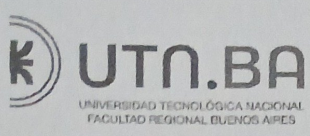


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Resuelto por Alexander Popartorn



ANÁLISIS MATEMÁTICO II
Examen Final 23/07/2024

APELLIDO DEL ALUMNO: NOMBRE:

CORRIGIÓ: REVISÓ:

T1	T2	P1	P2	P3	P4	CALIFICACIÓN

Todas las respuestas deben ser justificadas adecuadamente para ser tenidas en cuenta.

No resolver el examen en lápiz. Duración del examen: 2 horas

Condición de aprobación (6 puntos): tres ejercicios correctamente resueltos (uno de T1 o T2 y dos de P1, P2, P3 o P4).

T1) Indique si cada una de las siguientes proposiciones es verdadera o falsa. Fundamente correctamente la respuesta.

- a. El flujo del campo vectorial $\vec{F}(x, y, z) = (x + y, y + z, x + z)$ a través de la superficie S:
 $x^2 + y^2 + z^2 - 2z = 0$ con $z \geq 1$, orientada con la normal de componente z positiva, es igual a $\frac{\pi}{2}$.
- b. La ecuación del plano tangente a la superficie de ecuación $z = f(x, y) + x^2$ en el punto en $(1, 2, 4)$ es $z = 2x + 2$ sabiendo que, f es una función de clase $C^2(\mathbb{R}^2)$ y admite un valor máximo local 3 en el punto $(1, 2)$.

07

T2) a. Defina punto silla de un campo escalar $f: U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$.

b. Muestre que $f(x, y) = x^2 \ln(y - 5)$ admite en $(x_0, y_0) = (0, 6)$ un punto silla.

P1) Sea $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ dada por $f(x, y) = \begin{cases} \frac{x \cdot \sin y}{y} & \text{si } y > 0 \\ xy & \text{si } y \leq 0 \end{cases}$ Para cada $\vec{v} \in \mathbb{R}^2$ analice la existencia de las derivadas direccionales $f'((0, 0), \vec{v})$. Justifique la respuesta.

P2) Plantee la integral triple (sin calcular) que permite hallar del volumen del cuerpo definido por

$z \leq 4 - x - y$, $2x + y \geq 4$ en el primer octante.

P3) Sea $g: \mathbb{R}^3 \rightarrow \mathbb{R}$ una función de clase $C^2(\mathbb{R}^3)$ y el campo vectorial $\vec{F}(x, y, z) = (yz + 2, g(x, y, z), z)$.

Si se sabe que la circulación de $\vec{F}$ a lo largo de la curva C parametrizada por $\vec{r}: [0, \pi] \rightarrow \mathbb{R}^3$ / $\vec{r}(t) = (\cos t, \sin t, \sin t)$ es igual a 1, calcule el flujo del rotor de $\vec{F}$ a través de la porción del plano $S: y = z$, contenida en $x^2 + y^2 \leq 1$, $y \geq 0$ considerando la normal de componente z positiva. Grafique S y C .

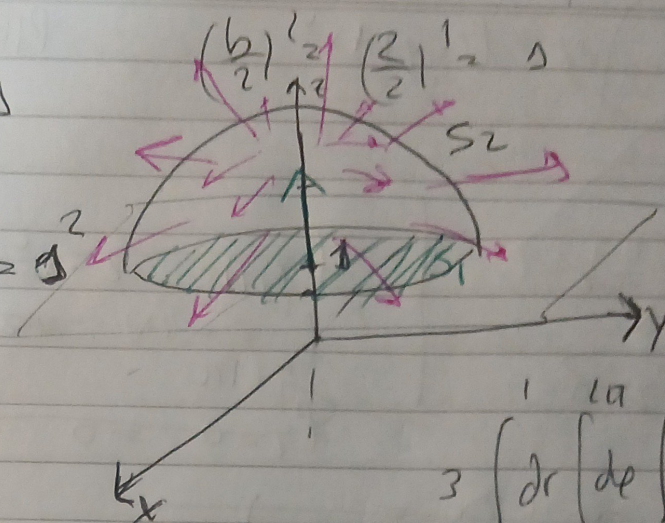
P4) Determine la solución de la ecuación diferencial $(2y - 4x^2)dx + xdy = 0$ que pasa por el punto $(1, 1)$.

T1

$$z^2 - 2z + 1 = 1$$

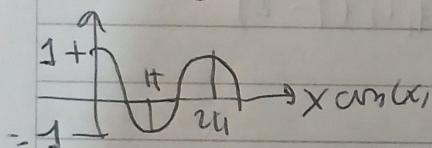
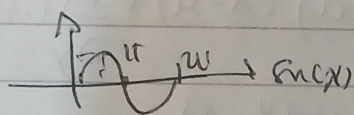
$$(z-1)^2 = 1$$

$$x^2 + y^2 + (z-1)^2 = 1$$



$$\iint \vec{F} \cdot d\vec{\sigma} = \iiint_V \text{div} \vec{F} \, dV$$

$$\text{div}(\vec{F}) = (1+1+1) = 3$$



$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$$

$$\int_0^{2\pi} \int_0^{\pi/2} \int_0^1 r^2 \sin \theta \, dr \, d\theta \, d\phi = 0 - (-r^2) = r^2$$

$$r^2 \phi \Big|_0^{2\pi} = \int_0^{2\pi} 2\pi r^2 \, d\phi = \frac{2\pi r^3}{3} \Big|_0^{2\pi}$$

$$+ (-\pi) = 2\pi = 3\pi = \iint \vec{F} \cdot d\vec{\sigma}$$

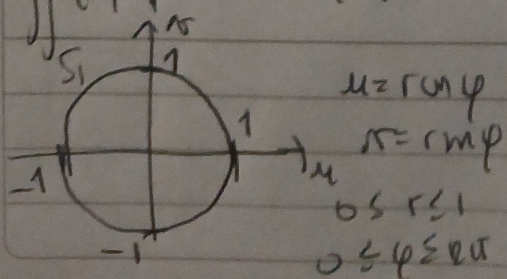
$$\iint_{S_1} \vec{F} \cdot d\vec{\sigma} + \iint_{S_2} \vec{F} \cdot d\vec{\sigma} = 2\pi$$

by calculation by Gauss' theorem

$$\sigma = \left(\frac{4\pi r^3}{3} \right) \cdot 3 = 4\pi r^3$$

$$\iint_{S_1} \vec{F} \cdot d\vec{\sigma} = \begin{vmatrix} x=u & \vec{T}(u,v) = (u,v,1) \\ y=v & \vec{T}'_u = (1,0,0) \\ z=1 & \vec{T}'_v = (0,1,0) \end{vmatrix} \vec{T}'_u \times \vec{T}'_v = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0,0,1)$$

$$\iint_{S_1} (x, y, u+1) \cdot (0,0,1) \, d\vec{\sigma} = \iint_{S_1} (u+1) \, du \, dv$$



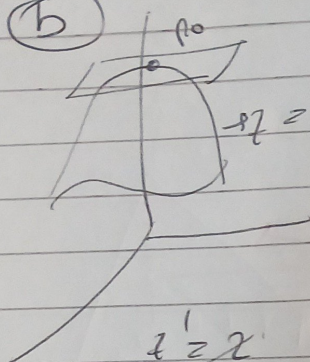
$$\begin{aligned} u &= r \cos \phi \\ v &= r \sin \phi \\ 0 &\leq r \leq 1 \\ 0 &\leq \phi \leq 2\pi \end{aligned}$$

$$\int_0^{2\pi} \int_0^1 (r \cos \phi + 1) r \, dr \, d\phi = \frac{r^2}{2} + \frac{r^2 \cos \phi}{2} \Big|_0^1 = \frac{1}{2} + \frac{\cos \phi}{2}$$

$$\int_0^{2\pi} \left(\frac{1}{2} + \frac{\cos \phi}{2} \right) d\phi = \frac{1}{2} \phi + \frac{\sin \phi}{2} \Big|_0^{2\pi} = \frac{1}{2} (2\pi) = \pi$$

$$z(x+1)$$

(b)



$$z = f(x, y) + x^2$$

$$P_{\text{plan}} = z = 2x + 2$$

$$\downarrow f'_x$$

$$z'_x = 2$$

$$x$$

$$z'_y = 0$$

$$f(x, y) = z - x^2$$

$$f'_x = z - 2x = -2$$

$$f'_y = 0$$

$$f'_x = 0 \Rightarrow 0 = 0$$

$$z'_x = f'_x + 2x = 1$$

$$P_0 = (1, 2, 4)$$

$$z'_y = f'_y + 0 = 0$$

$$z = f(x, y) + x^2$$

$$P_0 = (a, b, c)$$

$$\text{eq. da plano} \quad z = c + \left(\frac{\partial f(a, b)}{\partial x} + 2a \right) (x - a) + \frac{\partial f(a, b)}{\partial y} (y - b)$$

$$z = 4 + (f'_x(1, 2) + 2)(x - 1) + f'_y(1, 2)(y - 2)$$

mtb.

$$f(1, 2) = 3$$

$$f'_x(1, 2) = 0$$

$$f'_y(1, 2) = 0$$

$$z = f(1, 2) + 1^2 = 3 + 1 = 4$$

aval a (1, 2, 4)

$$z = 4 + (2)(x - 1) + 0 = 4 + 2x - 2$$

$$\boxed{z = 2x + 2} \quad \text{Verdade}$$

$$P=1$$

$$Q=x$$

$$P_4) (2y + 4x^2) dx + x dy = 0 \quad \int \frac{M'_y - N'_x}{N} dx = \int \frac{2 - 1}{x} dx$$

$$\begin{matrix} P'_y = 2 \\ Q'_x = 1 \end{matrix}$$

$$M = e^{\int \frac{1}{x} dx} = e^{\ln|x|} = e^{\ln x} = x$$

$$= e^{\int \frac{2-1}{x} dx} = e^{\ln x} = x$$

$$\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln|x|} = e^{\ln x} = x$$

$$\mu(x) = e^{\ln x} = x$$

$$\mu(x) = x$$

$$x(2y - 4x^2) dx + x^2 dy = 0 \rightarrow (2yx - 4x^3) dx + x^2 dy = 0$$

$$\begin{matrix} P'_y = 2x \\ Q'_x = 2x \end{matrix} \rightarrow V(x,y) = \int (2xy - 4x^3) dx = x^2y - x^4 + \phi(y)$$

$$V'_y = x^2 + \phi'(y) = x^2$$

$$\phi'(y) = 0 \rightarrow \phi(y) = 0$$

$$V(x,y) = x^2y - x^4 + C = h$$

$$S_h: x^2y - x^4 = h \rightarrow 1 - 1 = 0$$

$$x^2y - x^4 = 0 \rightarrow y = \frac{x^2}{1} = x^2$$

$$(y = x^2) \text{ SP}$$

Alexander Popov tutor

P1) Answer denot $\vec{v} = (a, b)$:

$$\vec{v} = (a, b) \Rightarrow a^2 + b^2 = 1 \quad \lim_{h \rightarrow 0} \frac{f((0,0) + h(\vec{v})) - f(0,0)}{h} = \frac{f(ha, hb) - f(0,0)}{h}$$

$$\lim_{h \rightarrow 0} \frac{f(ha, hb) + 0}{h} = \lim_{h \rightarrow 0} \frac{f(ha, hb)}{h}$$

$$a = \frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{2}}$$

$$hb > 0 \rightarrow \lim_{h \rightarrow 0} \frac{ha \cdot \sin(hb)}{hb} = \frac{ha \sin(hb)}{h^2 b} = \frac{\sin(hb)}{hb} \rightarrow a$$

$$hb \leq 0 \rightarrow \lim_{h \rightarrow 0} \frac{ha \cdot hb}{h} = ha \cdot b = 0$$

$$\nabla f'(0,0)(a,b) \neq 0$$

A direct answer exists for $\vec{v} = (a,b) \in \mathbb{R}^2$

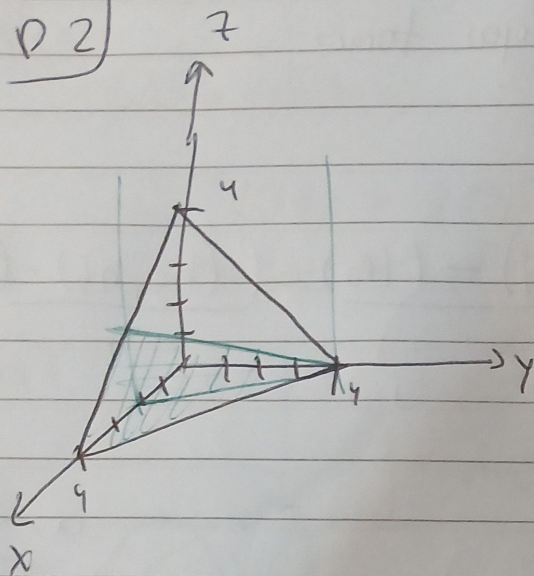
$$f'(0,0)(\vec{v}) = \begin{cases} a & \text{if } b > 0 \\ 0 & \text{if } b \leq 0 \end{cases}$$

Example:

$$\vec{v}(1,1) \rightarrow f'(0,0)(1,1) = 1 \quad b > 0 \quad (1 > 0)$$

$$\vec{v}(1,-1) \rightarrow f'(0,0)(1,-1) = 0 \quad b \leq 0 \quad (-1 < 0)$$

D2)



$$z = 4 - x - y$$

$$z = 0 \rightarrow z = 4 - y$$

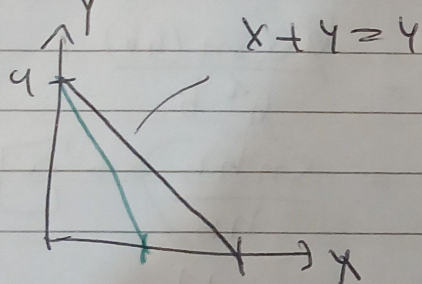
$$\frac{z}{4} + \frac{y}{4} = \frac{4}{4}$$

$$y = 0 \rightarrow z = 4 - x$$

$$\frac{z}{4} + \frac{x}{4} = \frac{4}{4}$$

$$(1) \quad z + x + y \leq 4 \quad (0,0,0) = 0 \leq 4$$

$$(2) \quad \frac{2x}{2} + \frac{y}{2} = 4 \rightarrow \frac{x}{2} + \frac{y}{2} = 2$$



$$(2) \quad \frac{x}{2} + \frac{y}{2} = 2 \quad (2)$$

$$x = 2 - \frac{1}{2}y$$

$$\int_0^4 dy \int_{2-\frac{1}{2}y}^{4-y} dz \int_0^{4-x-y} dx$$

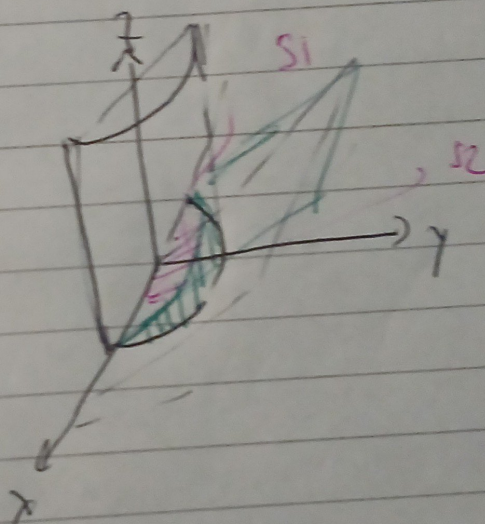
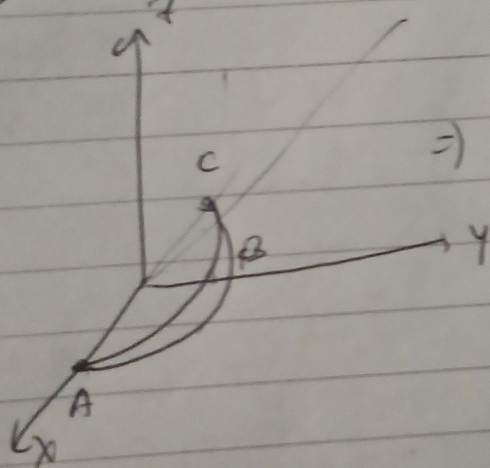
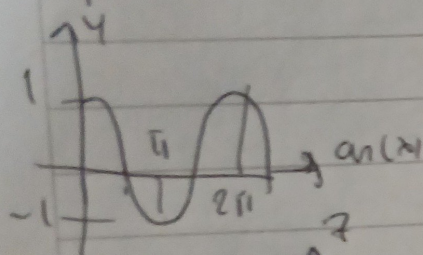
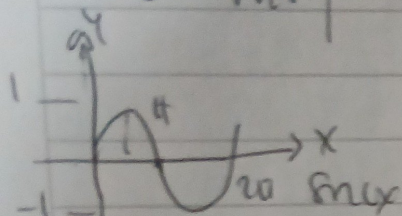
$$5 \quad x = \frac{4-y}{2} = 2 - \frac{1}{2}y$$

P3

$$\begin{array}{c|c|c|c} 0 & 1 & 1 & 1 \\ \hline x = \cos t & = 1 & -1 & 0 \\ y = \sin t & = 0 & 0 & 1 \\ z = \sin t & = 0 & 0 & 1 \end{array} \quad \iint_{\text{rot}} \text{rot}(\vec{F}) \cdot \vec{n} \, d\sigma =$$

$$\text{rot}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \begin{pmatrix} 0 - g'(x,y,z)(y_3) \\ - (0 - y)j + (g'(x,y,z)(y_3) - z)k \end{pmatrix}$$

$$\text{rot}(\vec{F}) = (-g'(x,y,z)(y_3), y, g'(x,y,z)(y_3) - z)$$



$$\iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma + \iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma =$$